CSI3007 – ADVANCED PYTHON PROGRAMMING

COGNITIVE GAMING & MEMORY ASSISTANCE PLATFORM FOR ELDERLY DEMENTIA PATIENTS

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1. Abstract

The Cognitive Gaming and Memory Assistance Platform is a Python-based supportive system designed to provide personalized and adaptive cognitive activities for elderly users experiencing memory and cognitive difficulties. The platform focuses on memory, attention, recognition and orientation. Literature from 2021–2026 shows growing research on serious games, computerized cognitive training, digital cognitive stimulation, smartphone/web applications and adaptive training for older adults and people with dementia or mild cognitive impairment. The proposed project integrates these established directions into one lightweight platform using Python and Streamlit. Caregivers can provide familiar information such as names, places, hobbies, routines and objects, which can be used to personalize activities. The system records accuracy, response time and attempts and applies explainable rule-based logic to adjust difficulty. English and Tamil accessibility and voice interaction are included to improve usability. A caregiver dashboard provides performance summaries and progress information. The platform is an academic prototype and supportive cognitive-assistance tool, not a medical diagnostic or treatment system.

2. Introduction

Dementia and age-related cognitive decline can affect memory, attention, recognition and orientation. Elderly users may also find generic digital applications difficult when the interface is complex, the language is unfamiliar or the activity difficulty does not match their ability. Digital cognitive training and serious games provide a way to deliver structured activities while collecting performance information.

The proposed project builds on this research by combining personalized cognitive activities, adaptive difficulty and caregiver monitoring in a single platform. Instead of using only generic questions or fixed games, caregiver-approved information about familiar people, places, objects, hobbies and routines can be incorporated into activities. The system is designed to be simple, accessible and suitable for Python-based academic implementation.

3. Problem Statement

- Generic cognitive games may not reflect the user's familiar memories and daily experiences.
- Fixed difficulty can make an activity too easy or too difficult for an individual user.
- Elderly users may face difficulties with complex interfaces, small controls or unfamiliar language.
- Caregivers need a simple way to review activity performance and progress.
- Existing digital cognitive solutions often concentrate on one intervention or technology instead of combining personalization, adaptation and monitoring.

4. Objectives

- Develop a Python-based cognitive gaming and memory assistance platform.
- Provide memory, attention, recognition and orientation activities.
- Use caregiver-provided familiar information to personalize activities.
- Record accuracy, response time, attempts and activity performance.
- Adjust activity difficulty using recent performance.
- Support English and Tamil interaction and voice features where feasible.
- Provide a caregiver dashboard for performance monitoring.
- Create a simple elderly-friendly interface suitable for academic demonstration.

5. Literature Review

A literature review of 20 relevant publications from 2021–2026 was compiled. The studies cover serious games, computerized cognitive training, digital cognitive stimulation, smartphone/web applications, adaptive training, usability and caregiver support. The table keeps each paper concise while showing the existing work, its relevance and what is new in the proposed project.

Year	Paper	Key contribution / Existing work	Relevance to project	What is new in our project
2021	An Individual Cognitive Stimulation Therapy App for People with Dementia and Carers: Results from a Feasibility Randomized Controlled Trial	Evaluated app-based individual cognitive stimulation for people with dementia and carers.	Supports digital cognitive stimulation and caregiver involvement.	Combines stimulation with caregiver-approved personal memories, adaptive difficulty and monitoring.
2021	Effects of Smartphone Application-Based Cognitive Training at Home on Cognition in Community-Dwelling Non-Demented Elderly Individuals: A Randomized Controlled Trial	Studied home-based smartphone cognitive training for older adults.	Supports accessible digital cognitive training.	Adds personalized content and performance-based adaptation.
2021	Effect of Cognitive Training in Fully Immersive Virtual Reality on Visuospatial Function and Frontal-Occipital Functional Connectivity in Predementia	Investigated immersive VR cognitive training for visuospatial function.	Supports technology-assisted cognitive training.	Uses a simpler Python/Streamlit interface without VR hardware.
2021	Cognitive Training With Fully Immersive Virtual Reality in Patients With Neurological and Psychiatric Disorders: A Systematic Review of Randomized Controlled Trials	Reviewed randomized trials using immersive VR cognitive training.	Shows the range of digital cognitive approaches.	Focuses on low-complexity elderly-friendly interaction and caregiver support.
2022	The Effectiveness and Safety of Serious Games for Improving Cognitive Abilities Among Elderly People With Cognitive Impairment: Systematic Review and Meta-Analysis	Reviewed serious games targeting overall cognitive abilities.	Supports game-based cognitive activities.	Integrates multiple cognitive domains with personalized information and adaptive progression.
2022	The Effectiveness of Serious Games in Improving Memory Among Older Adults With Cognitive Impairment: Systematic Review and Meta-Analysis	Focused on serious games for memory.	Directly supports memory-game design.	Uses familiar people, places, objects and routines in memory tasks.
2022	Effectiveness of Serious Games for Improving Executive Functions Among Older Adults With Cognitive Impairment: Systematic Review and Meta-Analysis	Reviewed serious games for executive functions.	Supports game-based cognitive stimulation.	Adds performance-based difficulty adjustment across activities.
2022	A Systematic Review and Meta-Analysis of Randomized Controlled Trials on the Effect of Serious Games on People With Dementia	Reviewed randomized evidence on serious games for dementia.	Supports the research basis for serious-game interventions.	Combines games, personalization, adaptive progression and caregiver feedback.
2022	A Systematic Literature Review and Meta-Analysis on Digital Health Interventions for People Living With Dementia and Mild Cognitive Impairment	Reviewed digital health interventions for dementia and MCI.	Supports the digital-health foundation.	Combines cognitive activities, personalization, language support and caregiver monitoring.
2022	Efficacy and Moderators of Virtual Reality for Cognitive Training in People with Dementia and Mild Cognitive Impairment: A Systematic Review and Meta-Analysis	Reviewed VR cognitive training and moderators.	Supports technology-assisted cognitive training.	Provides a lightweight alternative without specialized VR equipment.
2022	Promotors and Barriers to the Implementation and Adoption of Assistive Technology and Telecare for People With Dementia and Their Caregivers: A Systematic Review	Identified factors influencing assistive-technology adoption.	Supports simple, usable and caregiver-supported design.	Uses simple navigation, familiar content and caregiver monitoring.

2023	The Performance of Serious Games for Enhancing Attention in Cognitively Impaired Older Adults	Examined serious games for attention.	Supports attention-focused cognitive games.	Combines attention with memory, recognition and orientation.
2023	Serious Games for Learning Among Older Adults With Cognitive Impairment: Systematic Review and Meta-Analysis	Reviewed serious games for learning.	Supports meaningful game-based learning.	Uses familiar personal information in learning and recall activities.
2023	Researched Apps Used in Dementia Care for People Living With Dementia and Their Informal Caregivers: Systematic Review on App Features, Security, and Usability	Reviewed dementia-care apps for features, security and usability.	Supports usability and caregiver requirements.	Integrates caregiver profile management, games and performance tracking.
2024	Computerized Cognitive Training for Memory Functions in Mild Cognitive Impairment or Dementia: A Systematic Review and Meta-Analysis	Reviewed computerized memory training.	Supports computer-based memory training.	Adds adaptive memory activities using recent performance.
2024	Effects of Electronic Serious Games on Older Adults With Alzheimer's Disease and Mild Cognitive Impairment: Systematic Review With Meta-Analysis of Randomized Controlled Trials	Evaluated electronic serious games for Alzheimer's disease and MCI.	Supports electronic game-based cognitive intervention.	Adds personal memories, caregiver monitoring and adaptive progression.
2024	Cognitive Training With Older Adults Using Smartphone and Web-Based Applications: A Scoping Review	Reviewed smartphone/web cognitive training and highlighted heterogeneity and individual differences.	Supports personalized digital cognitive interventions.	Provides a customizable Python platform with English/Tamil and voice interaction.
2025	Cognitive Training With Adaptive Algorithm Improves Cognitive Ability in Older People With MCI	Investigated an adaptive algorithm for cognitive training.	Directly supports adaptive difficulty.	Extends adaptive logic across multiple activities using explainable rules.
2025	Efficacy of Digital Therapeutic Applications for Cognitive Training Among Older Adults With Mild Cognitive Impairment or Dementia: A Systematic Review and Network Meta-Analysis of Randomized Controlled Trials	Compared digital cognitive-training applications.	Supports the role of digital cognitive training.	Proposes a lightweight prototype centered on personalization and caregiver monitoring.
2026	Efficacy of Digital Cognitive Stimulation Therapy for People With Dementia: A Systematic Review and Meta-Analysis	Reviewed recent digital cognitive stimulation evidence.	Supports continued digital cognitive-stimulation research.	Extends stimulation with personal memory content, adaptive activities, Tamil support and caregiver monitoring.

5.1 Literature Synthesis and Research Gap

The reviewed literature establishes strong research interest in digital cognitive training, serious games, computerized cognitive training and digital cognitive stimulation. It also shows increasing attention to smartphone/web applications, usability, caregiver involvement and adaptive algorithms.

- Many studies focus on one cognitive domain or one specific digital intervention.
- Personalization using the user's own familiar people, places, objects and routines is not consistently integrated into cognitive games.
- Adaptive algorithms are increasingly studied, but a simple explainable rule-based approach can be appropriate for an initial academic prototype.
- Usability and accessibility remain important for older adults, motivating large controls, simple navigation and language support.
- Caregiver involvement is important, but combining personalization, adaptive difficulty and performance monitoring in one lightweight platform remains a useful integration goal.

The proposed contribution is therefore integration rather than claiming that cognitive games or adaptive training themselves are new: caregiver information → personalization → cognitive activity → performance collection → adaptive difficulty → next activity → caregiver monitoring.

6. Proposed Framework

The proposed framework uses a modular Python architecture. Streamlit provides the interface, SQLite stores profile and activity information, Pandas supports data processing, and Plotly can display performance trends. Recent activity results are used by the adaptive engine to determine the next suitable difficulty.

CAREGIVER INFORMATION
Profile • Language • Familiar people • Places • Hobbies • Routines • Objects

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PERSONALIZATION LAYER
Select or generate activities using familiar information

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COGNITIVE ACTIVITY LAYER
Memory • Attention • Recognition • Orientation

↓

PERFORMANCE ANALYSIS
Accuracy • Response time • Attempts • Recent trend

↓

ADAPTIVE DIFFICULTY ENGINE
Increase • Maintain • Decrease difficulty

↓

NEXT PERSONALIZED ACTIVITY

↓

SQLITE DATABASE + CAREGIVER DASHBOARD

6.1 Framework Modules

Layer	Module	Purpose
1	Caregiver & Profile	Collect language preference and familiar information approved by the caregiver.
2	Personalization	Convert profile information into suitable activity content and memory cues.
3	Cognitive Activities	Provide memory, attention, recognition and orientation activities.
4	Performance Tracking	Record score, accuracy, response time, attempts and history.
5	Adaptive Difficulty	Increase, maintain or decrease difficulty using recent performance rules.
6	Database	Store profiles, attempts and performance history using SQLite.
7	Caregiver Dashboard	Display progress, category performance and recent results.
8	Voice & Language	Provide English/Tamil text and voice interaction where feasible.

7. Core Processing Flow

1. Caregiver enters or updates user profile and familiar information.
2. System selects a cognitive activity based on the profile and current difficulty.
3. User completes the activity.
4. System calculates score, accuracy, response time and attempts.
5. Adaptive logic compares recent performance with predefined thresholds.
6. Difficulty is increased, maintained or decreased.
7. The result is stored in SQLite.
8. Caregiver dashboard displays performance trends and category results.

8. System Modules and Implementation Approach

Module	Implementation role
User/Profile Module	Stores caregiver-approved profile details, language preference and familiar information.
Cognitive Game Module	Provides memory, attention, recognition and orientation activities.
Personalization Module	Uses familiar people, places, objects, hobbies and routines to select activity content.
Performance Module	Calculates accuracy, score, response time and attempts.
Adaptive Difficulty Module	Uses explainable rule-based logic to select the next level.
Database Module	Uses SQLite for local storage.
Caregiver Dashboard	Displays history, category performance and progress charts.
Language & Voice Module	Provides English/Tamil labels and voice interaction where feasible.
Visualization Module	Uses Plotly/Streamlit charts for performance trends.

9. Technical Stack

Component	Technology
Programming	Python
Interface	Streamlit
Database	SQLite
Data processing	Pandas
Visualization	Plotly
Adaptive engine	Python rule-based logic
Speech-to-text	Whisper, where feasible
Text-to-speech	Python-compatible TTS solution
Languages	English and Tamil
Development	VS Code, Git and GitHub

10. Adaptive Difficulty Method

The initial prototype uses transparent rule-based adaptation. The engine considers recent accuracy, response time and errors to select the next level.

Performance condition	Decision	Purpose
High accuracy + fast responses	Increase difficulty	Provide a more challenging activity.
Moderate/stable performance	Maintain difficulty	Continue practice at the current level.
Low accuracy + repeated errors	Decrease difficulty	Reduce task complexity and support successful completion.

11. Expected Contributions

- A unified Python platform connecting cognitive activities, personalization, adaptation and caregiver monitoring.
- Use of caregiver-approved familiar information as input to cognitive activities.
- Explainable rule-based adaptive difficulty based on observable performance.
- English and Tamil accessibility for the intended user group.
- Simple elderly-friendly interaction with large controls and clear feedback.

- Performance history for caregiver review.
- A modular foundation that can later be extended with machine-learning-based personalization.

12. Expected Outcome

- Users can perform structured cognitive activities through a simple interface.
- Activities can reflect familiar personal information.
- The system can change difficulty based on recent performance.
- Caregivers can review performance through a dashboard.
- The platform can store activity history locally using SQLite.
- The project provides a practical Python implementation for academic demonstration.

13. Limitations

- The system is an academic prototype and not a medical diagnostic or treatment system.
- The initial adaptive mechanism is rule-based and requires further validation.
- Voice recognition and text-to-speech performance can vary with language and available libraries.
- Personalization depends on appropriate information entered by the caregiver.
- Clinical effectiveness cannot be concluded without formal evaluation.

14. Future Scope

- Introduce machine-learning-based personalized activity recommendation.
- Expand Tamil support and add additional Indian languages.
- Add more cognitive games and difficulty levels.
- Add secure cloud synchronization for authorized caregivers.
- Study longitudinal performance trends with larger datasets.
- Conduct formal usability studies with elderly users and caregivers.
- Explore validated clinical assessment integration only after appropriate expert and ethical evaluation.

15. Conclusion

The literature from 2021–2026 demonstrates substantial research on serious games, digital cognitive stimulation, computerized cognitive training, smartphone/web applications and adaptive cognitive training. These studies provide the foundation for the proposed platform. The project does not claim that these individual techniques are novel. Its proposed contribution is their integration into a simple Python-based system combining caregiver-approved personalization, multiple cognitive domains, explainable adaptive difficulty, English/Tamil accessibility and caregiver monitoring. The framework provides a clear path toward future usability evaluation, stronger personalization and validated adaptive models.

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